

Direct-Coupled Readers on a Reversible Field

Exact fibers, monotone forgetting, and the resolution at which a reference frame begins to pay

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Abstract

A reader coupled directly to a system — no estimator, no noise model, no belief about its own uncertainty — has nowhere for error to enter. Its output is an exact function of the state. This paper measures what such a reader can and cannot recover from a fully deterministic, exactly reversible integer field, and finds the structure sharper than the statistical version of the same question.

The substrate is a one-dimensional lattice gas: 128 cells, each carrying occupancy for a left- and a right-moving reader together with a four-bit payload, 1280 bits in total. Collisions mix payloads through a unimodular map over $\mathbb{Z}_{16}$, so two readers meeting compute rather than overwrite. Reversibility is exact: 11,146 collisions were run forward and then inverted, recovering every bit of the initial state. Nothing in the dynamics is stochastic.

Against this field a reader is a spatial window of s cells and a temporal window of τ steps. Because the coupling is direct, apparent stationarity is not a statistical statement: the field is exactly static to a reader precisely when the whole orbit lies inside a single fiber of that reader's map. Three regimes appear. For $s \leq 8$ the reader is saturated — it extracts $\log_2(T/\tau)$ bits, every distinction its sampling rate permits, and coarsening costs it nothing. Between $s = 16$ and 64 the field's correlation structure begins to cost it bits. At $s = 128$, $\tau = 256$ it extracts exactly zero: across the entire orbit the reading never changes once.

Coarsening is monotone. Every tested step of resolution loss, in space and in time, exactly and under quantization, merges fibers without re-cutting them, so the coarse reading is a strict function of the fine

one. Resolutions form a nested chain. Forgetting a layer therefore cannot disturb what lies beneath it — the reader loses distinctions, the field is untouched.

Two cost results follow, the second correcting the first. Priced as a full re-scan, coarse readings appear to cost more and return less — 32,768 cells touched for zero bits against one cell for twelve. That accounting is wrong: nothing re-scans. Priced as a delta stream, cost and yield are the same quantity, and the zero-bit reader emits no frames at all. But delta coding only beats re-scanning below a crossover near $s = 16$ to 32. Above it, the field changes more per step than the entire state costs to state, and a fine reader is better off looking again than remembering. Coarseness is what makes a persistent reference frame worth having.

Finally, keyframes. A transformation keyframe — rule, occupancy and seed, 1296 bits — regenerates every frame of the orbit, and a complete state snapshot of the same field costs 1280. The transformation keyframe is therefore a state keyframe plus 16 bits: generativity costs 1.25 per cent overhead on top of recording where the system is, and it pays from the second frame onward. The rule is 16 of those 1296 bits, so 79 per cent of what must be carried is initial condition, not law. And the trade is space for time, since the transformation yields frame t only after t steps — except where the rule is linear, in which case 128 steps collapse into a fixed matrix and arbitrary futures are reachable by squaring. The state at step 128×10^{12} was obtained in forty squarings and verified exactly against iteration at step 5,120. Whether a given system admits that collapse is undecidable in general, which is the precise sense in which a process owes an observer nothing beyond its own execution.

1. Direct coupling

A reader mediated by an estimator has two uncertainties: the error actually present in its measurement, and the error it believes is present. Results in that setting turn on whether the two agree. When they do, controller statistics can be invariant to the noise scale; when they diverge, the invariance breaks. That is the architecture of scale-invariant leakage, and it is a statement about a controller's beliefs.

This paper studies the case underneath it. A direct-coupled reader applies a fixed function to the state and returns the result. There is no estimator, no assumed variance, no ensemble. Consequently there is no gap in which error can live, and the questions change accordingly. One does not ask whether the reader is fooled; one asks what its fibers are. Apparent stationarity is not an illusion suffered by the reader but an exact property of the map: the field is static to a reader if and only if the orbit does not leave one preimage class.

The measurements below are of that kind throughout. Every number is exact, in the sense that it is a count over a deterministic orbit rather than an estimate over an ensemble.

2. The substrate

The field is a ring of $N = 128$ cells. Each cell may hold a right-moving and a left-moving reader, and carries two four-bit payloads, giving 10 bits per cell and 1280 bits in total. Occupancy is initialised once at random; the payloads are seeded from the first 128 hexadecimal digits of π , a choice of convenience that plays no role in any result.

Each step has two phases. Where a cell holds both a right- and a left-mover, the payloads mix:

$$(a, b) \rightarrow (a + b, a + 2b) \bmod 16$$

This map has determinant 1 over $\mathbb{Z}_{16}$ and is therefore invertible, with inverse $(a, b) \rightarrow (2a - b, b - a)$. Both outputs depend on both inputs, so the collision genuinely computes: it is a two-in, two-out operation, not a composition of two values into one. Streaming then advances right-movers by one cell and left-movers by minus one, which is a permutation of the ring.

The whole step is therefore a bijection on the state space, and reversibility was verified rather than assumed. Five hundred steps were run forward through 11,146 collisions and then inverted step by step; occupancy and both payload fields were recovered exactly, bit for bit. No information is destroyed by any collision.

This matters because the naive account of colliding readers is that they overwrite. **They do not have to.** A pointer map that sends many cells to one is lossy, but the field update built on it need not be: what collides are positions, not information. The distinction is measurable and the measurement is the reversibility check above.

One structural feature is worth stating early because it recurs. Occupancy in this system is rigid — right-movers rotate by +1 and left-movers by -1 every step, with period exactly 128 — and carries no

information about the evolving state. All 1280 bits of behaviour live in the payloads. The lattice is a solid bar; only the colours move.

3. Readers and fibers

A reader $R(s, \tau)$ averages the payload field over s adjacent cells and τ consecutive steps, then quantises at $\delta = 0.5$. Its output over an orbit is a sequence of readings, and the quantity measured throughout is the number of distinct readings it ever produces.

A note on what that logarithm is. $\log_2$ of the count is the log-cardinality of the reader's realised output alphabet over the tested orbit. It is an exact combinatorial quantity and it is what every table here reports. It is not, without further argument, the mutual information between the reader's output and the 1280-bit state: that would require a source distribution over states, and this paper works with a single deterministic orbit rather than an ensemble. The two coincide only under assumptions not made here. Where the text below says a reader 'extracts n bits' it should be read as the log-cardinality statement, which is the one that was measured.

A reader that produces one reading forever has a single fiber containing the whole orbit.

Table 1 gives that count over 4,096 steps.

$\tau \backslash s$	1	2	4	8	16	32	64	128
1	12.00	12.00	12.00	12.00	12.00	10.57	5.39	2.00
2	11.00	11.00	11.00	11.00	11.00	9.87	4.95	2.00
4	10.00	10.00	10.00	10.00	10.00	9.12	4.64	2.00
8	9.00	9.00	9.00	9.00	9.00	8.15	4.09	1.58
16	8.00	8.00	8.00	8.00	7.97	6.89	3.46	1.00
32	7.00	7.00	7.00	7.00	6.83	5.36	2.81	1.00
64	6.00	6.00	6.00	6.00	5.29	3.58	1.58	1.00
128	5.00	5.00	5.00	5.00	4.46	2.81	1.58	1.00
256	4.00	4.00	4.00	4.00	3.58	2.00	1.00	0.00

Table 1. $\log_2$ of the number of distinct readings over a 4,096-step orbit — the log-cardinality of the reader's output alphabet, not a mutual information. The sampling ceiling is $\log_2(4096/\tau)$.

Three regimes are visible and they are qualitative, not gradations.

Saturated. For every $s \leq 8$ the entry equals $\log_2(T/\tau)$ exactly — 12, 11, 10, 9 and so on down the column. No two readings ever coincide. The reader is limited entirely by how often it samples and not at all by the field, and coarsening from one cell to eight costs nothing whatever. Whatever structure the field has at those scales, this reader is above it.

Partial. From $s = 16$ the entries fall below the ceiling: 10.57 where 12 was available, 6.89 where 8 was. Here the field's own correlation structure begins to cost the reader distinctions. This is the only region where the answer depends jointly on the reader and the field rather than on one of them alone.

Fibered. At $s = 128$, $\tau = 256$ the reader extracts 0.00 bits. Its reading over the entire orbit never changes once. Underneath, the field is reconfiguring every step at near-maximal entropy — the per-cell payload standard deviation over time is 4.547 against 4.610 for uniform Z_{16} — and that reader's output stream is empty. Not compressed: empty. The field is not approximately stationary to it. It is exactly stationary, and will remain so for the life of the orbit.

4. Forgetting is monotone

If a reader loses resolution, does it merely stop distinguishing cases it used to distinguish, or can it come to distinguish cases it previously merged? The first is forgetting; the second would be scrambling, and it would mean that losing one layer of resolution corrupts access to another.

The test is whether the coarse reading is a function of the fine one across the orbit — equivalently, whether every fine fiber lies wholly inside one coarse fiber. Exact averaging trivially satisfies this, since a mean over $2s$ cells is the mean of two means over s cells, but the reader quantises, and rounding is not obviously compatible with nesting.

coarsening step	exact means	quantised, $\delta = 0.5$
space 1 $\rightarrow$ 2 cells	nested	nested
space 2 $\rightarrow$ 4 cells	nested	nested
space 4 $\rightarrow$ 8 cells	nested	nested
space 8 $\rightarrow$ 16 cells	nested	nested
space 16 $\rightarrow$ 32 cells	nested	nested
time 1 $\rightarrow$ 2 steps	nested	nested
time 2 $\rightarrow$ 4 steps	nested	nested
time 4 $\rightarrow$ 8 steps	nested	nested
space 8 $\rightarrow$ 16 at $\tau = 8$	nested	nested
time 8 $\rightarrow$ 16 at $s = 8$	nested	nested

Table 2. Every tested coarsening merges fibers without re-cutting them, including under quantisation.

Ten of ten, in both directions and at both exactness levels. Coarsening only merges. The reader's resolutions therefore form a chain of nested partitions over the field — a filtration — and moving down it is pure loss of distinction with no reorganisation.

The consequence is that Table 1 should not be read as a family of different readers. It is one reader at successive resolutions, and read down and to the right it is a forgetting curve. The field is identical in every cell of that table: same orbit, same 1280 bits, same trajectory. Only the fiber grows. What is lost is lost at the reader's end, and everything finer remains present in the object whether or not anything can still resolve it.

One boundary should be stated plainly. Nesting was verified for a reader family whose coarsening operation is averaging. A reader that coarsens by a hash, a threshold or a maximum carries no such guarantee, and a broken lattice there would mean that losing one layer genuinely does corrupt another. That case has not been tested here.

5. What a reading costs

The obvious accounting charges a reader for the cells it inspects. On that basis the finest reader touches one cell and returns twelve bits, while the coarsest touches 32,768 and returns none — four and a half orders of magnitude more work for strictly less information, monotone the whole way, with no crossover. It is a striking table and it is wrong.

Nothing re-scans. A reader that has already seen the field pays only for what changes, exactly as a video codec emits nothing for an unchanged block. Charging for change instead of for extent gives Table 3.

reader (s, τ)	blocks / frame	frames	blocks changed	delta bits / frame	bits extracted
(1, 1)	128	4096	93.3 %	1428.4	12.00
(8, 1)	16	4096	82.8 %	104.7	12.00
(32, 1)	4	4096	62.8 %	11.5	10.57
(128, 1)	1	4096	38.3 %	0.6	2.00
(8, 8)	16	512	85.4 %	101.9	9.00
(128, 8)	1	512	46.8 %	0.7	1.58
(64, 64)	2	64	27.0 %	1.1	1.58
(128, 64)	1	64	6.3 %	0.1	1.00
(32, 256)	4	16	15.0 %	1.8	2.00
(128, 256)	1	16	0.0 %	0.0	0.00

Table 3. Cost under delta coding against information returned. The two move together.

Cost and yield are the same quantity. The coarse reader is cheap for exactly the reason it is uninformative, and the last row makes the identity total: 0.0 per cent of blocks changed, 0.00 bits extracted, an output stream with no frames in it. Apparent stillness costs nothing to carry because

stillness and no-information are one measurement, not two. There is no penalty for abstraction and no bargain in it.

6. Where a reference frame begins to pay

Delta coding is cheap only relative to a reference. A stream of differences with no keyframe decodes to nothing. So the question is not whether to re-scan but whether carrying a reference and tracking changes beats simply looking again, and that has a crossover.

reader (s, τ)	keyframe (bits)	delta / frame	ratio	cheaper
(1, 1)	634	1428.4	2.25	re-scan
(2, 1)	317	637.8	2.01	re-scan
(4, 1)	157	258.0	1.64	re-scan
(8, 1)	73	104.7	1.43	re-scan
(16, 1)	34	36.7	1.08	re-scan
(32, 1)	15	11.5	0.78	carry deltas
(64, 1)	6	3.0	0.50	carry deltas
(128, 1)	2	0.6	0.30	carry deltas
(128, 64)	1	0.1	0.06	carry deltas
(128, 256)	1	0.0	0.00	carry deltas

Table 4. Above the crossover near $s = 16$ to 32 , the field changes more per step than the whole state costs to state.

At the finest resolution the field changes 1428 bits' worth per step against a 634-bit state. A reader there gains nothing from memory: re-scanning everything each instant is strictly cheaper than recording what moved. Below the crossover the ratio falls away and never returns.

So a reader does not avoid re-scanning because the world is still. It avoids re-scanning because it is coarse. Low resolution is what places an observer on the side of the crossover where a persistent reference frame is worth its cost. Sharpen the same reader past that point and change outruns encoding, and the only viable strategy is to look again, holding nothing. Coarseness is the precondition for memory paying for itself, not a deficiency memory compensates for.

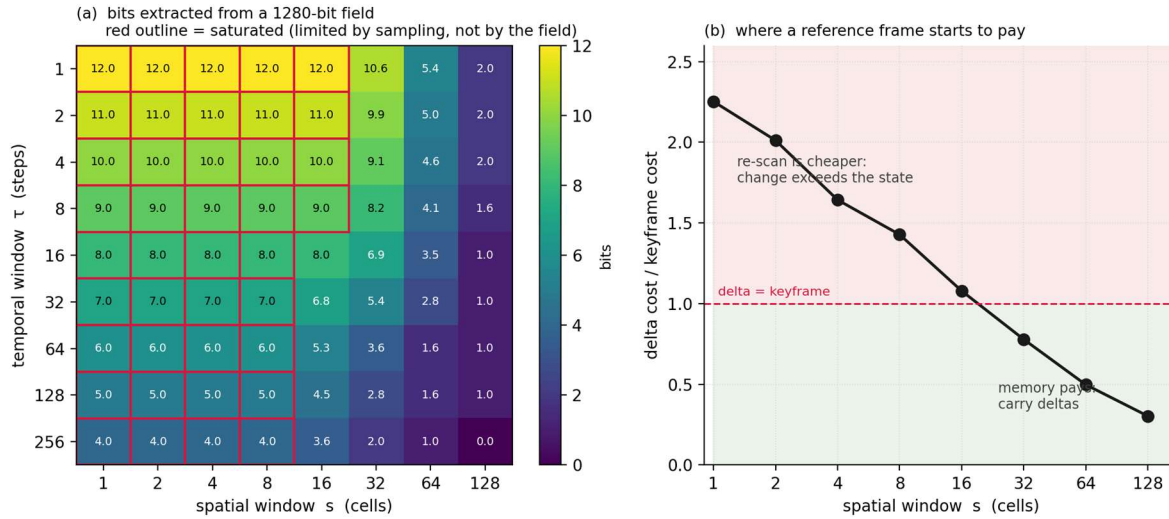


Figure 1. (a) Bits extracted as a function of the reader's window; red outlines mark cells at the sampling ceiling, where the reader is limited by its own rate rather than by the field. The bottom-right entry is the single-fiber case. (b) The ratio of delta cost to keyframe cost against spatial resolution, crossing unity between $s = 16$ and 32 .

7. Two kinds of keyframe

A reference frame can be a state or a rule. The two have different prices and different failure modes.

what is carried	bits	yields	break-even
occupancy pattern	256	—	—
payload seed	1024	—	—
full state keyframe (the two above)	1280	one frame	—
collision rule	16	—	—
transformation keyframe (state + rule)	1296	every frame	2 frames
reader's quantised keyframe, $\delta = 0.5$	634	one reading	3 frames

Table 5. The transformation keyframe is a full state keyframe plus the rule. Two break-even figures are given because the reader's quantised keyframe is a different object from a field snapshot.

The essential relation is additive rather than competitive. A transformation keyframe is not an alternative to recording the state; it is the state plus 16 bits. Generativity costs 1.25 per cent overhead on a snapshot one has to carry anyway, and from the second frame onward it wins and never stops winning — by a factor of about 4,000 over the 4,096-step orbit and without bound thereafter. Measured against the reader's quantised keyframe of 634 bits the crossover falls at three frames instead of two;

the two figures describe different objects and both are reported so neither is mistaken for the other. Two observations follow that are less obvious than the ratio.

The rule is 16 bits; the seed is 1024. Seventy-nine per cent of what must be carried is initial condition rather than law, and the rule is 1.25 per cent of a bare snapshot. If that proportion is at all general, then what an inheritance mostly transmits is a position, with a nearly free rule attached — the rule being shared across everything that runs the same dynamics, and the variation living in the seed.

And it is a trade of space for time. A state keyframe yields frame t immediately. A transformation keyframe yields frame t only after running t steps. This is the same trade the Bailey–Borwein–Plouffe formula makes for the digits of π : a program of a few hundred bits specifies every digit, uses logarithmic space, and charges time per digit actually wanted. Nothing is stored and everything is reachable, provided one is willing to wait. What that structure cannot do is random access, which is the entire reason a codec keeps keyframes at all.

8. Learning versus waiting

Waiting is not the only route to a distant state. If the dynamics can be analysed, the future may be computable without being traversed. This system permits exactly that, and it permits it for a reason that is easy to remove.

Because occupancy returns to its initial configuration every 128 steps and the collision is linear over Z_{16} , those 128 steps compose to a single fixed linear map M on the 256-dimensional payload space. The state after 128k steps is M^k applied to the seed, and M^k is reachable by repeated squaring.

target step	by iteration	by squaring	verified
$128 \times 40 = 5,120$	5,120 steps	6 squarings	exact match
128×10^3	10^5 steps	10 squarings	—
128×10^6	10^8 steps	20 squarings	—
128×10^{12}	10^{14} steps	40 squarings	—

Table 6. Analytic evaluation against iteration. The construction of M costs about two seconds, once.

The analytic and iterated states were compared element by element at step 5,120 and agree exactly. Beyond that, forty squarings reach a step count that no amount of iteration would.

Replace the collision with a nonlinear substitution and the construction above no longer applies: there is no fixed linear operator to raise to a power, and no shortcut of this kind is available. Same ring, same occupancy, same reversibility — one changed rule moves the system from analysable by this method to traversable-only by it. What this paper does not establish, and does not claim, is any general result about which systems admit such a collapse. Exhibiting the operator for a linear rule is a construction, not a decision procedure, and no procedure is offered here for determining in advance whether one exists.

8.1 An aside, outside the demonstrated result

The following is offered as an analogy to orient the construction above, not as a result of this paper. Its claims are about the published cryptanalytic literature and are cited accordingly; nothing in Sections 2 to 7 depends on them.

The engineered extreme of this contrast is a cryptographic hash. SHA-256 is fully disclosed — every constant, rotation and round is published — so nothing is hidden from analysis, while its search space forbids traversal outright. Waiting is not a slow option there; it is not an option. Every advance against it has therefore come from structure: preimage attacks reaching 43 of 64 steps by splitting the function into independently computable halves matched in a birthday phase, at around $2^{254.9}$ compression-function operations with memory of roughly 2^6 words [4]; the first practical collision on a 31-step reduction, at time complexity $2^{49.8}$ and memory 2^{48} , running in about 1.2 hours on 64 threads [5]; semi-free-start collisions at 39 steps. None extends to the full function. The design claim is not that an attacker must wait. It is that structural analysis must climb the whole distance and there is no partial credit. The correspondence with the linear-versus-nonlinear contrast above is suggestive and is not demonstrated here.

9. Status

Established here. Exact reversibility through 11,146 collisions. The three reader regimes, with the single-fiber case exhibited. Monotone coarsening in ten of ten tested steps, exact and quantised. The identity of cost and yield under delta coding. The keyframe crossover near $s = 16$ to 32. The additive relation between state and transformation keyframes: 1280 bits of snapshot plus a 16-bit rule, break-even at two frames on the full field and three against the reader's quantised keyframe. Exact analytic evaluation at step 5,120 and the squaring schedule beyond it.

Corrected within this work. An earlier draft compared a 1296-bit transformation keyframe against the reader's 634-bit quantised keyframe and reported break-even at two frames; those are different objects and the arithmetic gives three. Restating both on a full field snapshot of 1280 bits gives two, and exposes the additive relation above. Colliding readers were first treated as destroying information; the reversibility test refutes that, and the error was to measure the pointer graph rather than the field. The full-re-scan cost accounting, which produced an apparent inverse between work and information, is an artefact of charging for extent rather than for change.

Withdrawn. An earlier estimate that recurrence time grows at about 0.73 bits per added cell, fitted through non-monotonic data and extrapolated fourfold beyond its range. Holding the ring fixed at $N = 16$ and varying only the collision matrix moves the period from 16 to 1,038,352 — five orders of magnitude — so no scaling law in N alone is supportable. Periods appear governed by the operator's multiplicative order, with rules of equal order reproducing period columns exactly, though order alone does not determine them: two rules of order 8 differ by a factor of 63.

Not claimed. Nothing here concerns thermodynamics. That a finite reversible system recurs is a theorem about bijections on finite sets, and the recurrence observed in this toy is not evidence about

statistical mechanics; the connection to Boltzmann's reply to the recurrence objection is an interpretation and is not supported by anything measured. Nor does anything here bear on physics: the substrate is a one-dimensional lattice gas, and a one-dimensional lattice gas cannot produce Navier–Stokes behaviour for the same reason two-dimensional regularity is settled — there is no vorticity to stretch. Whether the lattice-gas correspondence survives requires at least two dimensions and sufficient rotational symmetry, which is why the square-lattice model of 1973 fails where the hexagonal model of 1986 succeeds.

Relation to prior work in this programme. Scale-invariant leakage concerns a reader mediated by an estimator, and its results turn on the match between true and believed uncertainty. The present work sits upstream of that: with direct coupling there is no belief and hence no mismatch, and apparent stationarity becomes a fiber property rather than a calibration property. The two are complementary rather than overlapping, and the earlier warning that invariant internal diagnostics can accompany growing absolute excursions applies to any use of Table 1 as a stability indicator.

10. Open

Non-averaging readers. Monotone forgetting was verified only for coarsening by averaging. Whether the fiber lattice survives coarsening by hashing, thresholding or maximum is untested, and a failure there would mean that losing one layer of resolution can corrupt access to another.

What governs the period. Operator order predicts period columns for rules of equal order but does not determine them. The finer invariant is unknown and is a well-posed question about the conjugacy structure of the collision map in $GL(2, \mathbb{Z}_{16})$ interacting with the ring length.

The partial regime. The middle band of Table 1, where the answer depends jointly on reader and field, is the only region carrying information about the field's correlation structure. Its width and position as a function of N and of the collision rule have not been measured.

Two dimensions. Everything here is one-dimensional. The reader results should transfer unchanged, but the correspondence with fluid behaviour requires a lattice with enough rotational symmetry, and that is where the substrate would have to go to connect with anything physical.

References

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[4] **Preimages for step-reduced SHA-2.** Advances in Cryptology — ASIACRYPT 2009, Lecture Notes in Computer Science. Meet-in-the-middle preimage attacks on 43-step SHA-256 and 46-step SHA-512.

[5] Liu, F. & Li, Y. (2024). The first practical collision for 31-step SHA-256. Advances in Cryptology — ASIACRYPT 2024. Best paper award.

References [4] and [5] support only the aside in Section 8.1 and are quoted from the published summaries of those results; readers relying on the exact complexities should consult the primary papers.

A reproducibility package accompanies this paper: the engine, the reversibility and nesting checks, and scripts regenerating every table and the figure from a fresh computation in under a minute on one core. All arithmetic is exact integer arithmetic over $\mathbb{Z}_{16}$; no floating-point tolerance enters any result except the reader's own quantisation, which is stated as $\delta = 0.5$ throughout.